Protective Egg Shell Design Milestones

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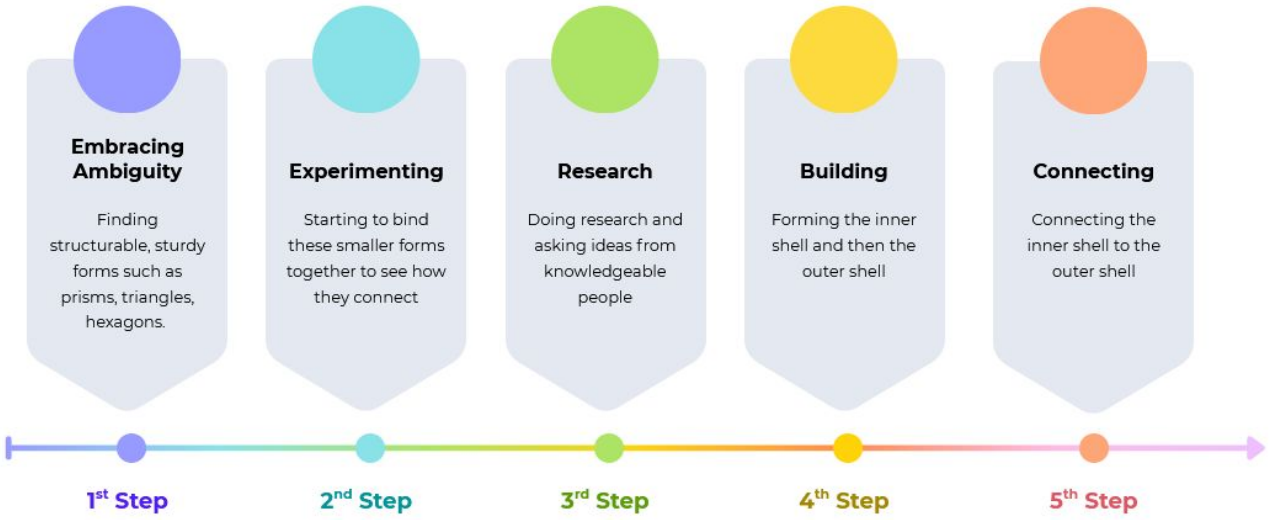
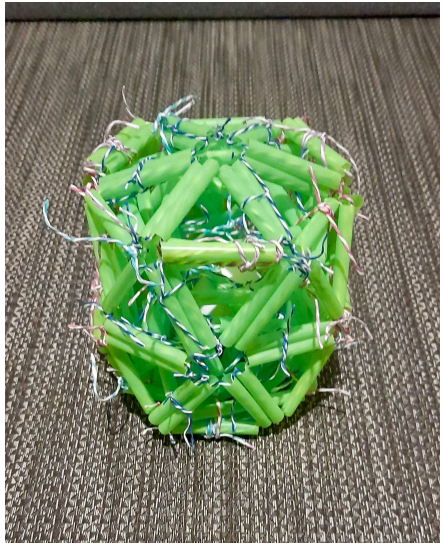
What I did for my First Prototype (No AI Involved)





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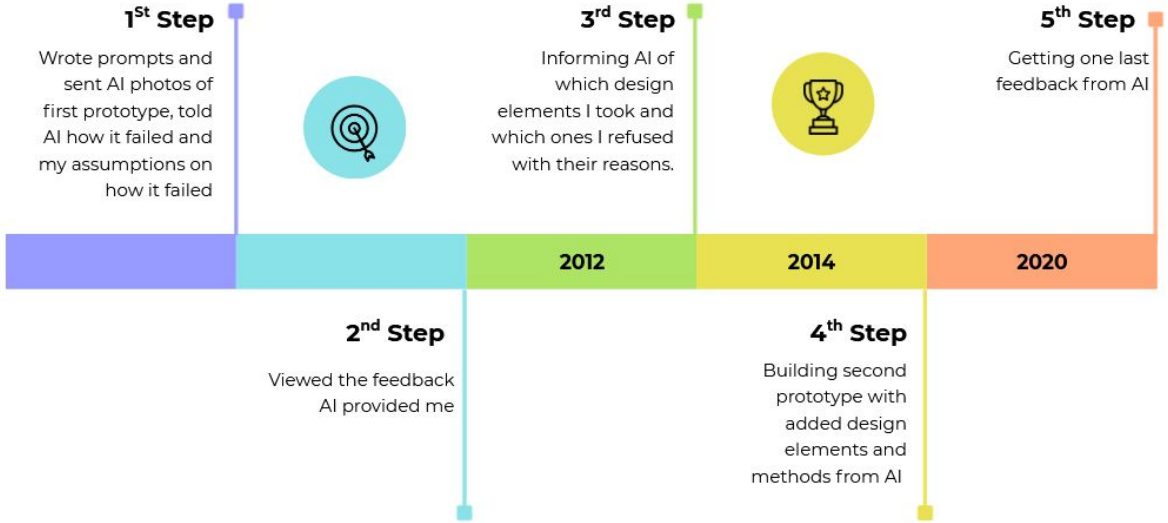
What I did for my Second Prototype (AI Involved)



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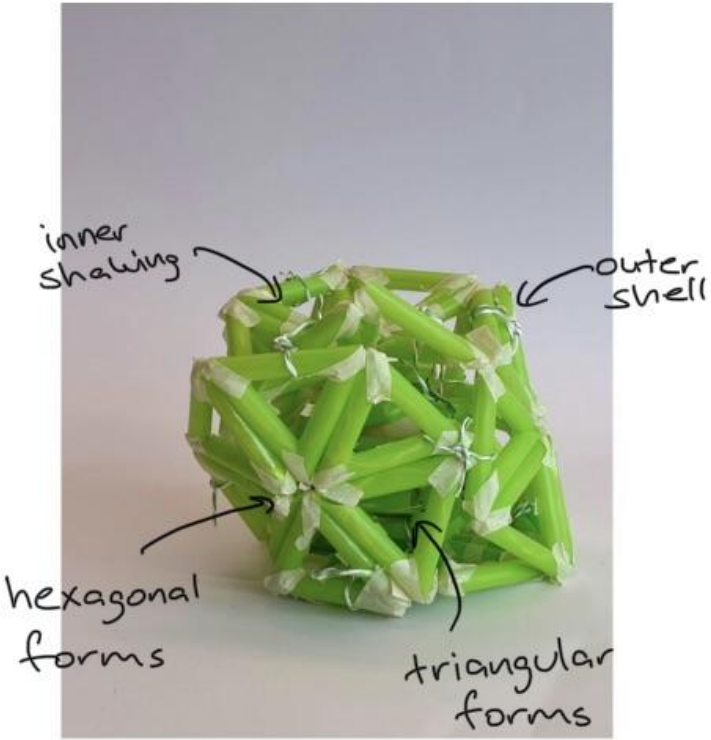
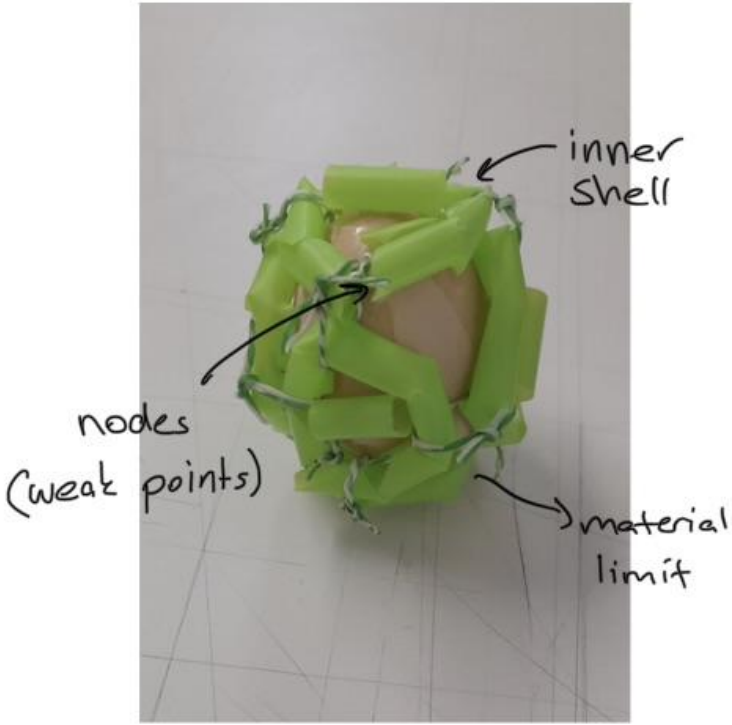
Keywords


inner shaking
outer shell
hexagonal forms
triangular forms


inner shell
nodes (weak points)
material limit

My Design Process with Gemini

Prompt:

“Hello, here I have provided you a pdf about my assignment about an egg drop challenge where I can only use plastic straws, tape and string. This will be my second prototype and I was thinking of doing two layers like shells to protect the egg. I need help with designing the first shell where it’ll be small hexagons surrounding the egg. Could you please visualize a few structured in this format?”

Attachments:

https://lms.ozyegin.edu.tr/pluginfile.php/156756/mod_assign/introattachment/0/Design%20Challenge_%20EggDrop%20%232.pdf?forcedownload=1

Gemini’s role here was to clarify the concept I had in mind.

Conceptualizing the Inner Hexagonal Shell

For the inner shell to protect the egg effectively, it should act as a rigid "exoskeleton." Here are three ways to structure those hexagons using your limited straw supply:

Design Concept	Structural Logic	Pro/Con
The Honeycomb Band	A single belt of 6 hexagons wrapped around the egg's "equator."	**Pro:** Uses very few straws. **Con:** Leaves the top and bottom vulnerable.

Design Concept	Structural Logic	Pro/Con
The Truncated Tetrahedron	A 3D shape where some faces are hexagonal, creating a "ball" of straws.	**Pro:** Excellent energy redirection. **Con:** Hard to tape securely.
The Hex-Suspension	Two hexagonal "rings" (top and bottom) connected by vertical struts.	**Pro:** Very stable center of gravity. **Con:** Higher risk of asymmetric collapse.

My Design Process with Gemini

The "Reality Check" (Senior Consultant Critique)

As your AI collaborator, I have to provide a "Reality Check" regarding your material constraints:

- **The Vertices Challenge:** Taping 6 straws together at a single point (a hexagonal vertex) is notoriously difficult with only 1 meter of tape.
- **The "Node" Failure:** In your reflection, you should note that straw ends are weak "nodes". If the hexagons are too small, you will have too many nodes, and the structure may fail at the joints before the straws even begin to compress.
- **Tension vs. Compression:** I suggest using the **string** to create "Tensegrity" within your hexagons. Use the straws for the hexagonal frame (compression) and the string to suspend the egg in the center (tension).

Gemini's role here is to act like a "Senior Consultant Critique" to comment on my design.

What I accepted? What I Rejected?

I accepted the recommendation of supporting the weak nodes pointed out by Gemini which were the ends of the straws.

But I refused to use strings to create "Tensegrity" because of my limited string length.

My Design Process with ChatGPT

Prompt:

“This was the outer shell of my first prototype. The egg didn’t hit the floor but it cracked from shaking inside the system. The inner layer was a triangle prism and I had added rectangle straws to cushion the egg. Also I’m thinking of cutting the straws into smaller pieces.”

Later on I prompted:

This is the inner shell. 6 hexagons in total. 4 surrounding the egg. One at the bottom one at the top

ChatGPT identified why my first prototyped failed and explained it to me in the role of a clarifier.

🔎 What Happened in Prototype 1

1️⃣ The Outer Shell

Your outer shell looks like a dense polyhedral cage (almost dodecahedron-like). That’s structurally strong in compression.

BUT…

It’s very rigid.

That means when it hit the ground:

- The structure stopped suddenly.
- The egg inside kept moving.
- The egg hit the internal supports.
- Crack = internal acceleration failure.

This is a **G-force problem**, not a floor impact problem.

You successfully prevented penetration.
You did not control deceleration.

My Design Process with ChatGPT

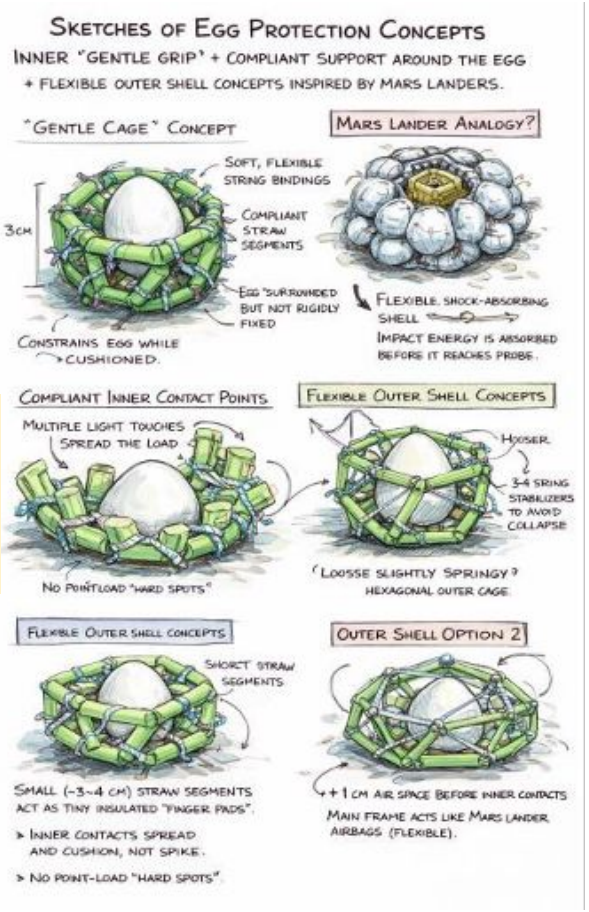
Prompt:

“Could you please draw me a few different sketches so I could understand visually?”

I rejected these sketches because they were not in the dual shell concept not only I but Gemini had also thought about. For me the top of the egg’s being unprotected made no sense.

However the visualizations of how to weave the triangle straws helped me a lot to create a flexible outer shell. It also gave me different ideas to form the straws.

ChatGPT expanded upon the egg protection concepts



My Reflection

I used ChatGPT and Gemini and tasked it to act like a “Senior Design Consultant” to refine a second prototype made with limited materials (straws, tape, and string). After many iterative prompts and feedback, both of them helped me analyze structural weaknesses in the first prototype—the internal shaking that cracked the egg in my first prototype

They suggested a dual-shell strategy: a rigid hexagonal inner cage to stabilize the egg and a flexible triangular outer shell to absorb impact through a crumple zone and tension-based suspension using string.

AI also critiqued the node stability, material constraints, and energy dissipation strategies.

My design failed because I couldn’t construct a stable modul with a limited amount of materials. 1st prototype had a good outer shell.

The 2nd prototype had a good inner shell.

In the end Gemini was helpful for understanding the assignment and improving structural decisions, though it focused more on solving the task than on generating broader design ideas. ChatGPT was not acting like a “Senior Design Consultant” but came up with more ideas then Gemini.